

Residual-Calibrated Multivariate Forecasting and Stress Testing under Distributional Change

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Abstract

Reliable financial forecasts require more than accurate conditional means. They also require predictive distributions that represent joint uncertainty, tail risk, and the possibility that market behavior changes after a model is fitted. This study develops an interpretable framework for calibrated and robust multivariate financial forecasting. A vector autoregression (VAR) captures predictable interactions among a small set of related asset returns, while competing joint residual models represent the uncertainty left unexplained by the VAR. Gaussian, multivariate Student- t , empirical bootstrap, and diffusion-based innovation specifications are compared within the same conditional-mean model. Sampled innovations are propagated recursively to construct probabilistic return, price, and portfolio paths.

The framework evaluates whether forecast probabilities are credible before treating flexibility as evidence of robustness. Marginal, multivariate, and tail calibration are assessed alongside proper scoring rules and sharpness. Innovation-based stress tests then alter volatility, dependence, downside tails, and contamination to measure how forecast calibration and portfolio-risk estimates deteriorate under distributional change. Controlled simulations establish behavior under known innovation laws, while a rolling financial case study evaluates external validity. Diffusion-Innovation VAR (DI-VAR) is treated as a flexible candidate rather than a presumed solution: the objective is to determine which residual models produce reliable forecasts, under which conditions, and at what cost. The methodology provides a foundation for future work on residual-based shift detection, adaptive uncertainty models, and distributionally robust forecasting.

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1 Introduction

Multivariate financial forecasting has two distinct tasks. The first is to estimate which components of future asset behavior are predictable from recent cross-asset dynamics. The second is to characterize the uncertainty that remains after those dynamics have been modeled. Point forecasts address only the first task; risk management, interval estimation, and portfolio decisions require the second.

Vector autoregressions provide a transparent model of lagged interactions among financial variables. Their coefficient matrices can be inspected, stability can be diagnosed, and forecast paths can be generated recursively. Yet the usual Gaussian innovation assumption is difficult to reconcile with financial returns that exhibit heavy tails, asymmetric losses, volatility clustering, nonlinear dependence, and stronger co-movement during market stress. A VAR may therefore provide an adequate conditional-mean forecast while producing an unreliable predictive distribution.

This study separates conditional-mean modeling from uncertainty modeling. The VAR captures predictable cross-asset structure, and its rolling forecast errors represent what the model failed to explain. Alternative joint innovation models are fitted to those errors and propagated through the same VAR. This controlled comparison asks whether additional distributional flexibility produces forecasts that are not only sharper, but also better calibrated on chronologically later observations.

Diffusion models are included as a flexible method for learning joint residual distributions. The resulting Diffusion-Innovation VAR (DI-VAR) does not replace the VAR or directly model prices; it replaces the conventional parametric innovation law with samples from a learned denoising process. DI-VAR is compared with simpler Gaussian, Student- t , and empirical bootstrap alternatives. This hierarchy is essential because low-dimensional financial residuals may not contain enough information to justify a complex generative model.

Robustness is distinguished from flexibility. A model that closely reproduces historical residuals may still fail when volatility, dependence, or tail behavior changes. The framework therefore evaluates nominal calibration first and subsequently exposes every innovation model to the same controlled stresses. Robustness is measured by the rate at which calibration, proper scores, and portfolio-risk estimates deteriorate as the forecast environment moves away from its reference distribution.

The central research question is:

How does the choice of joint residual model affect the calibration and robustness of multivariate autoregressive forecasts under changes in volatility, dependence, and tail behavior?

The methodological sub-question is:

Under what residual structures and distributional shifts does diffusion-based innovation modeling provide meaningful improvements over Gaussian, Student- t , and empirical alternatives?

The contribution is a modular methodology connecting interpretable dynamics, joint uncertainty modeling, calibration, and distributional stress. The study remains informative if a simpler innovation model is preferable.

2 Conceptual Framework

The methodology contains four layers:

1. **Predictable dynamics.** A VAR estimates lagged interactions among related asset returns.
2. **Joint residual uncertainty.** Competing innovation models represent multivariate behavior left unexplained by the VAR.
3. **Calibration.** Forecast distributions are tested for marginal, joint, and tail reliability together with sharpness.
4. **Stress and robustness.** Controlled changes to the innovation law reveal how statistical credibility and portfolio-risk estimates deteriorate.

This decomposition gives each component an interpretable function. VAR coefficients describe mean transmission across assets. Residual diagnostics identify shape and dependence omitted by the mean model. Innovation-model comparisons determine whether extra distributional complexity is useful. Calibration determines whether reported probabilities correspond to observed frequencies. Stress-response curves identify failure thresholds and translate them into financial consequences.

3 Statistical Methodology

3.1 Financial representation

Let $P_t \in \mathbb{R}_+^K$ denote adjusted prices for K related assets. The default modeled variable is the vector of log returns,

$$y_t = \log P_t - \log P_{t-1}, \quad (1)$$

where the logarithm is applied componentwise. Returns are preferred to raw prices because a standard VAR requires approximately stationary inputs. If long-run equilibrium relationships among price levels are substantively important, a vector error-correction model is a separate extension.

The analysis uses a chronological training, calibration, and test split. The training period estimates conditional-mean dynamics. The calibration period supplies pseudo-out-of-sample forecast errors and model-selection information. The final test period remains untouched until the forecasting procedure is fixed.

3.2 Vector autoregressive dynamics

For $y_t \in \mathbb{R}^K$, a VAR of order p is

$$y_t = c + \sum_{\ell=1}^p A_\ell y_{t-\ell} + u_t, \quad (2)$$

where $c \in \mathbb{R}^K$, $A_\ell \in \mathbb{R}^{K \times K}$, and $u_t \in \mathbb{R}^K$ is the innovation vector. Lag order is selected using training or calibration observations, and the fitted system is checked for stability.

The VAR is intentionally common across innovation specifications. Holding the conditional mean fixed isolates the effect of uncertainty modeling. More complex forecasting systems can be considered in later robustness checks but are not required for the primary question.

3.3 Rolling forecast residuals

Purely in-sample residuals can understate deployment error. Innovation models are therefore trained on rolling one-step-ahead errors. At each calibration time t , the VAR is fitted or updated using information available through $t - 1$, and

$$\hat{u}_t = y_t - \hat{y}_{t|t-1} \quad (3)$$

is recorded. Each residual remains a joint K -dimensional observation so contemporaneous dependence is preserved.

If pronounced conditional heteroskedasticity remains, an auxiliary scale model may decompose

$$u_t = D_t z_t, \quad (4)$$

where D_t contains time-varying marginal scales and the innovation model learns standardized shocks z_t . Reporting results with and without standardization distinguishes flexible distribution learning from volatility forecasting.

3.4 Competing joint innovation models

Gaussian-VAR. The conventional benchmark assumes

$$u_t \sim \mathcal{N}(0, \hat{\Sigma}). \quad (5)$$

It is interpretable and efficient when the Gaussian assumption is adequate.

Student- t -VAR. A multivariate Student- t distribution provides a parsimonious heavy-tail model,

$$u_t \sim t_\nu(0, \hat{\Sigma}). \quad (6)$$

Its degrees-of-freedom parameter provides an interpretable measure of tail thickness.

Bootstrap-VAR. Joint residual vectors are sampled from the empirical residual distribution. A block-bootstrap variant preserves short-range temporal dependence. This is a strong low-dimensional nonparametric baseline.

Diffusion-Innovation VAR. DI-VAR fits a denoising diffusion model to joint residual vectors. For a clean residual u_0 , the forward process is

$$u_s = \sqrt{\bar{\alpha}_s} u_0 + \sqrt{1 - \bar{\alpha}_s} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (7)$$

A network $\epsilon_\theta(u_s, s)$ is trained through

$$\mathcal{L}(\theta) = \mathbb{E}_{u_0, s, \epsilon} \left[\|\epsilon - \epsilon_\theta(u_s, s)\|_2^2 \right]. \quad (8)$$

The learned reverse process generates joint innovations without imposing a Gaussian or elliptical form. Because the target dimension is small, model capacity is deliberately limited and performance is reported relative to computational cost and sample size.

3.5 Recursive probabilistic forecasts

For innovation model j , draw M future innovation paths. Each is propagated recursively through the fitted VAR:

$$y_{t+h}^{(m,j)} = \hat{c} + \sum_{\ell=1}^p \hat{A}_{\ell} y_{t+h-\ell}^{(m,j)} + \tilde{u}_{t+h}^{(m,j)}. \quad (9)$$

Return paths can be mapped back to prices through

$$P_{t+h}^{(m,j)} = P_t \odot \exp \left(\sum_{r=1}^h y_{t+r}^{(m,j)} \right), \quad (10)$$

where $\odot$ denotes componentwise multiplication. For portfolio weights w , every simulation also produces a portfolio-return or wealth path, allowing errors to be evaluated in decision-relevant units.

4 Calibration and Forecast Evaluation

Calibration requires predicted probabilities to agree with observed frequencies across repeated forecast origins. It is evaluated with sharpness because arbitrarily wide intervals can attain high coverage without being useful.

4.1 Marginal evaluation

For every asset and forecast horizon, the study reports prediction-interval coverage, width, interval score, quantile loss, probability integral transform diagnostics, expected calibration error, and marginal Continuous Ranked Probability Score. Horizon-specific results are retained rather than reporting only pooled averages.

4.2 Joint evaluation

Marginal calibration does not imply that simultaneous outcomes are represented correctly. Joint evaluation therefore includes the Energy Score, dependence recovery, multivariate rank diagnostics where practical, and frequencies of simultaneous downside events. These quantities target the cross-sectional structure that joint innovation modeling is intended to preserve.

4.3 Tail and portfolio evaluation

Financial usefulness is assessed through lower-tail coverage, Value at Risk exceedance rates, Expected Shortfall, portfolio-loss distributions, and maximum drawdown. An innovation model can appear satisfactory at the asset level while materially understating joint portfolio losses.

4.4 Comparative protocol

Each innovation method uses the same VAR fits, forecast origins, horizons, and Monte Carlo sample counts. Results include mean performance, uncertainty across origins and random seeds, computational cost, and failure cases. DI-VAR is considered useful only when calibration or robustness gains are material relative to simpler alternatives.

5 Robustness and Distributional Stress

Robustness is the ability to retain useful probabilistic performance when the deployment innovation law differs from the reference law. The nominal model estimates a reference distribution $\hat{P}$ from calibration residuals. Stress operators generate a family Q_γ indexed by severity γ .

5.1 Stress scenarios

Volatility stress. Innovation scale is inflated according to

$$u^{\text{stress}} = \gamma u, \quad \gamma > 1. \quad (11)$$

Dependence stress. The innovation dependence structure is modified to increase common movement or downside correlation while controlling marginal scales. This tests diversification failure during market declines.

Tail stress. The frequency or magnitude of adverse innovations is increased through tail reweighting, contamination, or asymmetric transformations.

Asset-specific and outlier stress. A shock is applied to one component and propagated through the fitted system, or a controlled fraction of extreme joint residuals is introduced.

Historical stress. Residual patterns from turbulent periods are used as interpretable scenarios, subject to the limitations of small historical samples.

Wasserstein stress. Distributional ambiguity is summarized by

$$\mathcal{Q}_\varepsilon = \left\{ Q : W(Q, \hat{P}) \leq \varepsilon \right\}. \quad (12)$$

For a financial or calibration loss L , a robust risk summary is

$$\sup_{Q \in \mathcal{Q}_\varepsilon} \mathbb{E}_Q[L(Y)]. \quad (13)$$

The loss may represent portfolio loss, drawdown, tail exceedance, or calibration error. The Wasserstein radius is reported as a sensitivity parameter rather than assigned an unsupported universal interpretation.

5.2 Robustness curves

For each model and stress family, degradation relative to nominal performance is

$$\Delta_{m,j}(\gamma) = m_j(Q_\gamma) - m_j(\hat{P}), \quad (14)$$

where m is an error or loss metric. Robustness curves reveal whether a model fails gradually, abruptly, or only under particular shifts. This is more informative than declaring a model robust from nominal results.

6 Empirical Design

6.1 Financial case study

The primary application uses approximately four liquid and economically related assets. Asset selection is justified before comparative results are observed. Adjusted prices are aligned to common trading dates and converted to daily or weekly log returns. The analysis records missing-data decisions, corporate-action adjustments, sampling frequency, and portfolio weights.

Evaluation uses rolling forecast origins. At each origin, model fitting and innovation estimation use only prior observations. Hyperparameters are selected using training and calibration data, after which forecasts are evaluated on chronologically later outcomes. Multiple market regimes and test windows are included where sample size permits.

6.2 Controlled simulation study

Simulations complement the financial study by providing known innovation laws. Stable multivariate VAR systems are generated under Gaussian, Student- t , mixture, and heteroskedastic regimes. Every innovation model is evaluated under every regime and across repeated random seeds. These experiments test recovery of known moments, quantiles, dependence, and forecast coverage; the financial case study evaluates external validity.

6.3 Ablation studies

The planned ablations include:

- joint versus independently fitted innovation models;
- in-sample versus rolling out-of-sample residuals;
- raw versus volatility-standardized residuals;
- ordinary versus block residual bootstrap;
- residual sample size and diffusion capacity;
- nominal versus stressed evaluation;
- asset-level versus portfolio-level calibration.

These comparisons identify whether apparent gains arise from diffusion, joint modeling, volatility treatment, or experimental leakage.

7 Hypotheses

1. Gaussian-VAR is competitive when residuals are approximately elliptical and stable.
2. Student- t -VAR improves tail calibration when heavy tails dominate but dependence remains relatively simple.
3. Bootstrap-VAR is competitive in low dimensions when sufficient representative residuals are available.
4. DI-VAR is most valuable when joint residual structure is non-elliptical or multimodal and the calibration sample supports flexible estimation.

5. No unconditional innovation model remains calibrated under every conditional volatility or regime shift.
6. Dependence shifts can degrade portfolio-risk calibration even when marginal asset coverage remains acceptable.

8 Interpretation and Limitations

The framework is interpretable because it separates mean dynamics, residual uncertainty, calibration, and stress response. Nevertheless, VAR coefficients are predictive associations rather than causal effects. Financial returns may contain time-varying conditional means and variances that an unconditional residual model cannot represent. Rolling residual samples may be too small for stable diffusion training. Backtests reuse a finite and historically specific sequence of market conditions. Stress scenarios are sensitivity analyses rather than literal forecasts of future crises.

Most importantly, richer historical fit does not establish robustness. Diffusion may reproduce the reference residual distribution closely while extrapolating poorly under shift. Conclusions must therefore depend on out-of-sample calibration and controlled degradation, not visual sample quality alone.

9 Expected Contribution

The project contributes a common evaluation framework rather than a preferred algorithm:

- an interpretable decomposition of cross-asset forecasting and residual uncertainty;
- a leakage-aware rolling protocol for fitting joint innovation models;
- a fair comparison of parametric, empirical, and diffusion innovations;
- calibration diagnostics spanning marginal, joint, tail, and portfolio behavior;
- stress-response curves under volatility, dependence, tail, contamination, and Wasserstein perturbations;
- evidence describing when additional model complexity is warranted.

10 Future Research

The residual construction provides a direct path toward distributional-shift research. A monitoring system can compare recent residual windows with the reference distribution using Wasserstein distances, covariance and tail-dependence changes, two-sample statistics, or rolling calibration errors. Subsequent work can distinguish innovation shift from changes in VAR dynamics and determine whether to update the innovation model, refit the conditional mean, or widen robust uncertainty sets.

A further extension replaces the unconditional innovation law with

$$p_{\theta}(u_{t+1} \mid \mathcal{F}_t, z_t), \tag{15}$$

where z_t may encode recent volatility, drawdown, liquidity, or a latent regime. This conditional formulation could support adaptive probabilistic forecasting, but it requires more data and separate

evaluation of conditional calibration. The current framework supplies the baseline against which that complexity should be judged.

11 Conclusion

This study reframes multivariate financial forecasting as a sequence of distinct questions. A VAR asks what can be predicted from recent cross-asset behavior. A joint residual model asks what uncertainty remains. Calibration asks whether that uncertainty is credible. Stress testing asks how quickly credibility and financial risk estimates deteriorate when the innovation distribution changes.

Within this framework, DI-VAR is a flexible candidate rather than a guaranteed solution. Comparing it with Gaussian, Student- t , and bootstrap alternatives makes the analysis interpretable, falsifiable, and useful even when simpler models prevail. The methodology connects probabilistic forecasting with distributional shift and robust risk evaluation while preserving clear boundaries between predictive dynamics, uncertainty representation, and stress response.

A Planned Outputs

The final study will include data and split summaries, VAR diagnostics, residual-distribution diagnostics, nominal calibration tables, proper scores, tail and portfolio metrics, computational comparisons, ablation results, and stress degradation summaries. Planned figures include return and volatility diagnostics, residual pair plots, PIT and reliability diagrams, forecast fan charts, joint downside diagnostics, portfolio loss distributions, and robustness curves over stress severity.

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